

Quiz 8

Name: _____

Date: _____

1. Is $\tan x$ even, odd, or neither? explain why.

Hint: Use

$$\sin(-\theta) = -\sin \theta, \quad \cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = \frac{-\sin \theta}{\cos \theta} = -\tan \theta \quad \text{odd}$$

2. If $\sin \theta = \frac{1}{2}$ and θ is in Quadrant I, find $\cos \theta$.

Use

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \left(\frac{1}{2}\right)^2 + \cos^2 \theta &= 1 \end{aligned}$$

$$\begin{aligned} \frac{1}{4} + \cos^2 \theta &= 1 \\ \cos^2 \theta &= \frac{3}{4} \end{aligned}$$

$$\begin{aligned} \cos \theta &= \pm \frac{\sqrt{3}}{2} \\ \text{Since } \theta \text{ is in Quadrant I} \\ \text{so } \cos \theta &= \frac{\sqrt{3}}{2} \end{aligned}$$

3. Sketch one period of the graph of

$$y = \sin x$$

from $x = 0$ to $x = 2\pi$.

Label the five key points.

